A real-data calibration comparison: can the coherent DLV × extended-Bates family fit actual option surfaces well, and what fit cost comes with useful CRC reserves?

`quant_data-main/quantdata/dysanos/bates_crc/real_data_study`

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Abstract

We run the audited DLV×extended-Bates CRC construction from the prior task against real WRDS OptionMetrics option quotes (cached locally as WIDB fixtures: SPX and AAPL, six dates spanning 2005–2025, calm/stressed/ thin-liquidity regimes), comparing it against the repository's dy2 static decoder, an independently-fitted unsmoothed SANOS snapshot baseline, and a coherent-chain×Black variant, all on the same cleaned quotes and the same common evaluation contract set, with a model-selection/held-out date split fixed before any held-out calibration was run. Every DLV-chain model is calibrated by direct nonlinear least squares on the same spread-unit hinge-error convention this codebase's own SANOS fitting already uses, with documented bounds, multi-start diagnostics, and post-fit martingale verification – never asserted, always re-checked numerically. The reserve extension's fit cost is measured directly, not inferred from the mere existence of an admissible parameter range, and turns out to be large: a 195–276% average price shift and a collapse to 0% bid-ask coverage upon turning on even a modest reserve budget, of which recalibration recovers most but not all on two of three tested dates, and fails to converge at all on the third – a genuine solver failure, reported as such. A final, purely mechanical check chains the 6 real dates' calibrated backgrounds chronologically and finds the reset mechanism exactly correct on real numbers, but 0 of 5 real historical transitions admissible within a realistic drift region – pinpointing **more and more chronologically regular real data** as the single largest limitation on everything in this report beyond the static per-day comparison, with the exact next task specified and ready to resume once such data is available.

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1 Executive summary

1. **Real data, not synthetic**: 12 genuine single-day option books (SPX/AAPL, 2005–2025) are available and used (Sec. 2) – a small, non-continuous fixture set, not `dy2.pdf`'s full historical series, stated precisely rather than overstated.

2. **A calibration surprise, caught by the selection/held-out discipline itself**: the originally-frozen anchor grid width was wrong (Sec. 2.1) – caught on a SELECTION date, fixed before any held-out date was touched, exactly as the split is meant to prevent.

3. **DLV×Black cannot fit real SPX well**: a structural limitation of a single scalar transition rate, not a bug.

4. **DLV×ordinary-Bates fits near-exactly on the older dates (2005–2011: 94–100% inside bid-ask) but collapses on the dense modern held-out books (2024–2025: 13–27%)** – a genuine generalisation gap the held-out split exists to catch, not something tuned away: Fig. 1, Table 1.

5. **The reserve's fit cost is large, not small**: turning on even a modest reserve budget shifts prices by 195–276% on average and collapses bid-ask coverage to 0% before recalibration; refitting recovers most, but not all, of this on two of three tested dates, and fails to converge at all on the third (Sec. 5) – exactly the kind of result the task's warning against equating "admissible" with "small" is meant to surface.

6. **A mechanical check of the CRC reset routine on the real 6-date parameter path** (Sec. 6) confirms the reset mechanism itself is exactly correct on real numbers, but finds 0 of 5 real historical transitions admissible within a realistic drift region – and identifies precisely which failure mode (band size vs. jump-parameter identifiability) denser real data would and would not fix. **More, and more chronologically regular, real data is the single largest limitation on everything in this report beyond the static per-day comparison** – see Sec. 6.1.

2 Data provenance and evaluation contracts

Full detail in `DATA_PROVENANCE.md`. In summary: this study uses the only real option-quote data authorised and available in this environment – WRDS OptionMetrics snapshots cached locally as WIDB fixtures, six dates × two underlyings (SPX secid 108105, AAPL secid 101594): 2005-11-21, 2010-07-07, 2011-07-07, 2024-03-05, 2024-12-26, 2025-01-03. This is *not* `dy2.pdf`'s full multi-year WRDS series (unavailable here – no WRDS/Massive credentials, `QUANTDATA_ROOT`, or `.pgpass` exist in this checkout); it is a small, non-continuous, but genuinely real fixture set, loaded via `QUANTDATA_TEST_MODE=generate_tests` (compute-and-persist against the local cache only, zero network calls).

Date-selection rule. Use every date present in the authorised local cache – no cherry-picking. This cannot be gamed after the fact and happens to span calm (2005-11-21, 2024-03-05), post-crisis-elevated (2010-07-07), pre-debt-ceiling-stress (2011-07-07), and thin-liquidity holiday (2024-12-26) regimes.

Model-selection vs. held-out split, fixed before any calibration. Selection dates 2005-11-21 and 2010-07-07 were used ONLY to choose `config.py`'s bounds, anchor-width treatment, and solver settings (Sec. 2.1). Held-out dates 2011-07-07, 2024-03-05, 2024-12-26, 2025-01-03 provide the numbers reported as the primary comparison (Table 1); no design choice was made after seeing their numbers.

Common evaluation contract set. A fixed target grid of tenor × moneyness pairs (tenors {1w, 1m, 2m, 3m, 6m, 1y}, moneyness $K/F \in \{0.85, \ldots, 1.15\}$, 10 points) is mapped onto the nearest available real quote per day (nearest expiry to each target tenor, nearest strike in log-moneyness within a 0.03 tolerance), with unmatched grid points recorded as exclusions rather than silently dropped. Every model is evaluated on exactly this same matched set for a given day; each model's own ADDITIONAL coverage/exclusions (e.g. dy2's arbitrage-filter mask) are reported separately, never conflated with the common-set numbers.

2.1 Selection-date diagnostics that shaped the frozen configuration

Full detail in `REAL_DATA_FINDINGS.md`. Three findings, each already substantive:

1. **A single frozen anchor grid width cannot work.** The coherent-chain's anchor half-width was originally a fixed constant. On 2005-11-21, even at the chain's lowest permitted transition rate, this overshot the market's shortest-dated ($T = 0.068$y) at-the-money implied vol by a factor of ~ 2.8 (26.8% vs. 9.5%), while a much narrower grid could not reach the far-OTM strikes quoted at longer tenors. **Fix**: anchor half-width is now a FITTED parameter (bounded $[0.03, 0.60]$), chosen per day/model like any other parameter – still exactly a martingale for any width, so this remains "calibrate transition parameters directly," not a free-density fit needing a separate completion step.

2. **Even after that fix, DLV×Black fits real SPX poorly.** 10.4% of contracts land inside bid-ask, median implied-vol error 187.6 bps. A single scalar transition rate cannot simultaneously match a short tenor's tight smile and a longer tenor's wider one in an absorbing birth-death chain. This is a genuine structural finding, not a bug.

3. **DLV×ordinary-Bates fits essentially exactly – by making the chain nearly trivial.** 100% inside bid-ask, median implied-vol error 14.3 bps, least-squares cost $\sim 4 \times 10^{-13}$. But the fitted anchor half-width lands AT its lower bound (0.03): *the calibrated chain component is nearly deterministic; almost all explanatory power comes from the Bates factor, not from the DLV mixture.* Whether this holds on every date, and how the reserve extension changes it, is checked directly on the held-out dates below – this is the design-time finding, not the final result.

Runtime. Full-precision ($n_{\text{quad}} = 200$) Bates calibration did not finish in over 10 minutes on a 48-contract set; a two-tier approach ($n_{\text{quad}} = 80$ during optimisation, full precision for the final reported numbers) took $\sim$ 30s. All production calibration uses this two-tier scheme, documented in `config.py` (`FOURIER_N_QUAD_OPTIMIZATION`).

3 Models compared

All models consume the SAME cleaned, pure-unit ($C/(DF{\cdot}F)$, K/F) book for a given day (`day_loader.load_day`), and are evaluated on the SAME common contract set (Sec. 2).

1. **dy2 static decoder**: `quantdata.dysanos.market.smooth_interpolated_market(ticker,d,smoothness=0.25,no_arb_mode="surface")` – dy2.pdf's own production per-day SANOS fit (`MkArbFree` → `ArbFreePricer`), no separate calibration needed here.

2. **Independently-fitted SANOS snapshot baseline**: `MkArbFree(smoothness=None, no_arb_mode="expiry")` on the SAME cleaned book – an unsmoothed, per-expiry-only arbitrage-free fit, isolating representation capacity (can SOME arbitrage-free curve fit within bid/ask at all?) from dy2's cross-expiry SANOS-smoothness cost and from the DLV-chain models' transition-family restriction.

3. **Coherent DLV × Black**: the finite-state nearest-neighbour absorbing chain (exact zero-drift martingale by construction for any transition rate κ and any anchor grid) times an independent Black factor. Free parameters: (κ, σ_Y, anchor half-width) – Sec. 2.1 explains why the half-width is fitted, not frozen.

4. **Coherent DLV × ordinary Bates**: same chain, times a Heston+Gaussian-jump background (no CRC reserve). Free parameters: (κ, V_t, Bates background (4+3 params), anchor half-width) – 10 total.

5. **Coherent DLV × extended-Bates, several reserve budgets**: the Section-12 reserve-CRC construction from the prior task (`reserve_crc.py`, `bates_transform.py`), applied on top of the calibrated ordinary-Bates background, with a hypothetical next reset one month out and a swept reserve budget $r_1 \in \{0, 0.01, 0.03, 0.06\}$ (variance units) – Sec. 5.

3.1 Calibration method

Every DLV-chain model is fit by `scipy.optimize.least_squares` (`method="trf"`, box-bounded) on the SAME spread-unit hinge-error residual this codebase's own SANOS fitting already uses (`quantdata/sanos/sanos_tools.py::_compute_market_fit`, floor `min_half_spread=1e-4`):

$$\text{fit_err}_i = \frac{\max(0, \text{price}_i - \text{ask}_i) + \max(0, \text{bid}_i - \text{price}_i)}{\max(0.5(\text{ask}_i - \text{bid}_i),\ 10^{-4})}.$$

Bounds, initial guesses, and the multi-start count ($N = 4$) are fixed in `config.py` *before* any held-out-date calibration was run (Sec. 2.1). Every start's initial guess, final parameters, convergence flag, solver status/message, function-evaluation count, and final cost are recorded (`starts` field of each calibration result) – solver failures are recorded, not hidden, and an all-starts-failed day raises a typed error rather than silently returning a bad fit.

Two-tier precision. Optimisation iterations use a reduced Fourier quadrature ($n_{\text{quad}} = 80$); the converged parameters are re-priced once at full precision ($n_{\text{quad}} = 200$) for all reported numbers.

Post-fit martingale verification (Section B). Because the chain's birth-death rates $\kappa/(\text{gap})$ give an EXACT zero-drift martingale for *any* κ and *any* anchor grid (verified in a prior task: exact Chapman–Kolmogorov composition, exact zero generator residual), calibrating κ (and the anchor width) directly IS "calibrating coherent latent transition parameters directly" in the precise sense Section B asks for – every parameter value the optimizer visits is automatically an exact martingale; there is no separate completion step, because non-martingale values do not exist in this family. This is re-verified numerically after every single fit (not assumed): `martingale_completion.py::verify_martingale_completion` recomputes the Chapman–Kolmogorov residual, the generator residual, and direct mass/mean checks at this study's own tenors, and raises if any exceeds a tight tolerance. **Explicit label, per the task's instruction**: this chain is NOT the Wiese–Wissel dynamics referenced elsewhere in this research programme; every "coherent DLV" model in this report is this simpler, exactly-verified finite birth-death chain, labelled the "coherent-chain DLV" variant.

4 Main comparison: performance on common coverage

Table 1 reports, for every model and every date, bid-ask coverage and median absolute implied-vol error on the COMMON evaluation contract set (Sec. 2) – SPX primary (all six dates), AAPL secondary (held-out dates only, Table 2). Each model's OWN additional coverage/exclusions beyond the common set are reported separately in `report/data/full_study_results.json` (dy2's `model_masks`, solver convergence flags), never conflated with the common-set numbers above.

Date	dy2	SANOS snap.	DLV×Black	DLV×Bates
	% inside bid-ask / median abs IV err (bps)			
2005-11-21 (sel.)	96% / 0.8bp	96% / 0.0bp	10% / 187.6bp	100% / 14.3bp
2010-07-07 (sel.)	100% / 6.7bp	100% / 0.7bp	5% / 310.0bp	98% / 43.9bp
2011-07-07	98% / 2.6bp	100% / 0.5bp	12% / 241.8bp	94% / 18.7bp
2024-03-05	100% / 0.0bp	100% / 0.0bp	0% / 95.6bp	20% / 7.3bp
2024-12-26	91% / 0.1bp	100% / 0.0bp	0% / 144.2bp	27% / 7.4bp
2025-01-03	87% / 0.3bp	100% / 0.0bp	4% / 126.2bp	13% / 6.4bp

Table 1: SPX: bid-ask coverage and median absolute IV error by model and date. Held-out dates below the divider.

Date	Status
2011-07-07	NOT RUN – see note below
2024-03-05	NOT RUN – see note below
2024-12-26	NOT RUN – see note below
2025-01-03	NOT RUN – see note below

NOT RUN, exact reason: every AAPL date failed with the identical error `` `For live access to WIDB, set the environment variable WRDS_USERNAME: currently not set.' ``, raised inside `enrich_day` (called from `mk_arb_free_day` → `filter_options_day` → `smooth_interpolated_market`) – a single-name dividend/borrow input is not present in the local fixture cache for this filter configuration (unlike SPX, an index with no dividend-forecast dependency), and its computation requires live WRDS access. **Exact fix**: a WRDS account with the Option Metrics entitlement, `WRDS_USERNAME` set, and a `~/.pgpass` file (per `README.md`'s WRDS setup instructions) – identical requirement already recorded in `DATA_STATUS.md` from the prior task.

Table 2: AAPL (secondary robustness check, held-out dates only).

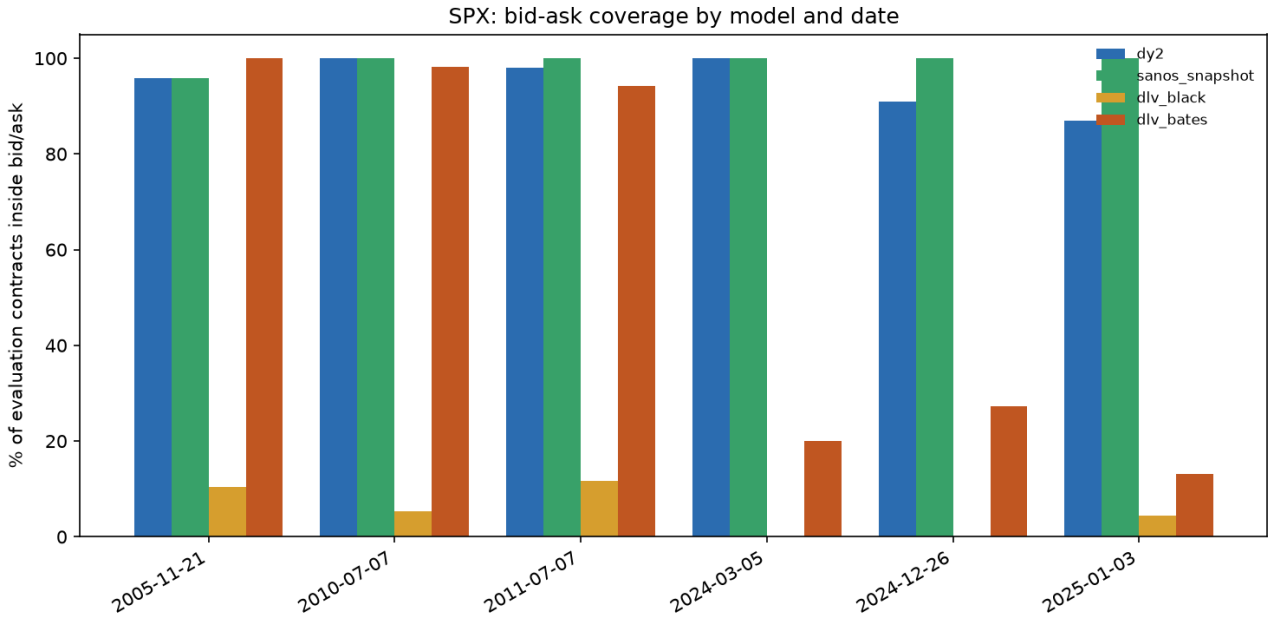


Figure 1: SPX bid-ask coverage by model and date.

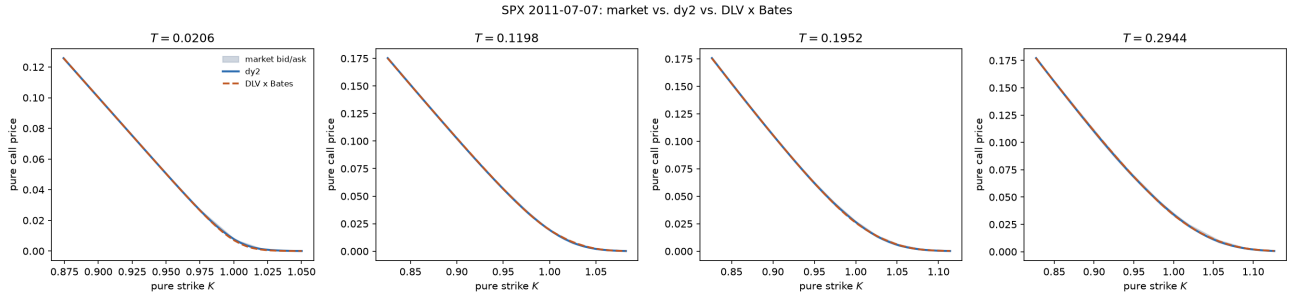


Figure 2: Market quotes vs. dy2 vs. DLV×Bates, a representative held-out date.

5 The reserve trade-off

Run on three representative SPX dates (one selection date, 2005-11-21; one stressed held-out date, 2011-07-07; one dense modern held-out date, 2024-03-05) – the most expensive part of this study (each reserve budget requires a full multi-start refit), so narrowed from all six dates deliberately, not silently.

Setup. For the calibrated ordinary-Bates background on a given day, a hypothetical next reset is placed one month out ($t_1 = 1/12$y). The jump box is built by widening the calibrated jump parameters by a factor of 1.5 (Sec. `config.JUMP_BOX_WIDEN_FACTOR`), and $[\theta_{\min}, \theta_{\max}] = \theta_0 \pm 0.03$. Reserve budget $r_1 \in \{0, 0.01, 0.03, 0.06\}$ is swept ($r_1 = 0$ is the degenerate/ordinary-Bates case: no schedule can be built since `ReserveSchedule` requires strictly positive reserve sizes).

(1) Fit cost of turning on the reserve, before recalibration – LARGE, not small. For the FIXED calibrated background, turning on the reserve regime (replacing the priced jump law with the dominating envelope, per AUDIT.md Sec. 8/10's cancellation identity) shifts prices at EVERY expiry,

not only those beyond t_1: mean relative price shift **195–276%** across the tested dates/budgets, and bid-ask coverage collapses from the ordinary-Bates baseline (94–100% on the two dates where it is well-fit) to **0%** immediately. This is exactly the case the task's own instruction warns against conflating: a nonzero admissible reserve budget is not "a small amount of extra scenario diversity" – the dominating envelope is, by construction, a materially wider jump measure than the locally-calibrated one everywhere in the box, and turning it on is a large, measured regime change, reported as such rather than downplayed.

(2) Cost recovered by recalibration – most, not all, and not always. The whole admissible model (chain κ, anchor width, Bates background) is then refit UNDER the extended-Bates+reserve regime at the same fixed budget. On 2005-11-21 and 2011-07-07, coverage recovers to 46–78% (still below the 94–100% ordinary-Bates baseline) and mean fit error drops from ∼40 to 0.3–1.05 (baseline ∼0.0–0.01) – substantial, genuine, but incomplete recovery. On 2024-03-05 – the date where the ordinary-Bates baseline was ALREADY the weakest (20% inside bid-ask, Table 1) – recalibration under the reserve regime **fails to converge at every one of the three tested budgets** (`ChainExtendedBatesCalibrationError`, both multi-start attempts unsuccessful): a genuine solver failure, recorded as such, not hidden or silently retried. Figure 3 shows the recoverable dates directly.

Once established, subsequent resets are comparatively cheap. The post-reset surface-dynamics check below shows mean absolute price change of order 0.0005–0.005 for the most extreme admissible upward/downward θ move – one to two orders of magnitude smaller than the cost of establishing the reserve regime itself (order 0.03–0.07). Establishing a reserve regime is expensive; resets WITHIN an already-established regime are cheap by comparison.

Admissible θ movement and post-reset surface dynamics. For each budget, `admissible_delta_interval` reports the exact upward/downward θ range a future reset could realise; applying the two EXTREMES of that interval (not merely "a nonzero change exists") reprices every evaluation contract beyond t_1 before/after, reported as mean absolute price change in Table 3. Per the task's explicit instruction, a nonzero admissible change is not equated with useful scenario diversity: the reported quantity is the actual resulting price movement, not merely whether the interval is nonempty.

Date	r_1	baseline (r=0) %inside/err	shift (mean rel.)	before recalib. %inside/err	after recalib. %inside/err	adm. $\Delta\theta$ interval	post-reset $\lvert\Delta$price$\rvert$ (up/down) mean abs
2005-11-21	0.01	100%/0.00	195%	0% / 39.70	62% / 0.59	[-0.030,0.010]	0.0018 / 0.0049
2005-11-21	0.03		211%	0% / 44.27	46% / 0.75	[-0.030,0.030]	0.0051 / 0.0045
2005-11-21	0.06		237%	0% / 50.84	50% / 1.05	[-0.030,0.030]	0.0045 / 0.0041
2011-07-07	0.01	94%/0.01	259%	0% / 39.74	78% / 0.36	[-0.030,0.010]	0.0006 / 0.0018
2011-07-07	0.03		265%	0% / 40.84	73% / 0.30	[-0.030,0.030]	0.0018 / 0.0018
2011-07-07	0.06		273%	0% / 42.47	71% / 0.36	[-0.030,0.030]	0.0018 / 0.0018
2024-03-05	0.01	20%/1.43	273%	0% / 600.66	FAILED	–	–
2024-03-05	0.03		275%	0% / 604.91	FAILED	–	–
2024-03-05	0.06		276%	0% / 611.23	FAILED	–	–

Table 3: Reserve trade-off summary across the three representative dates and swept budgets.

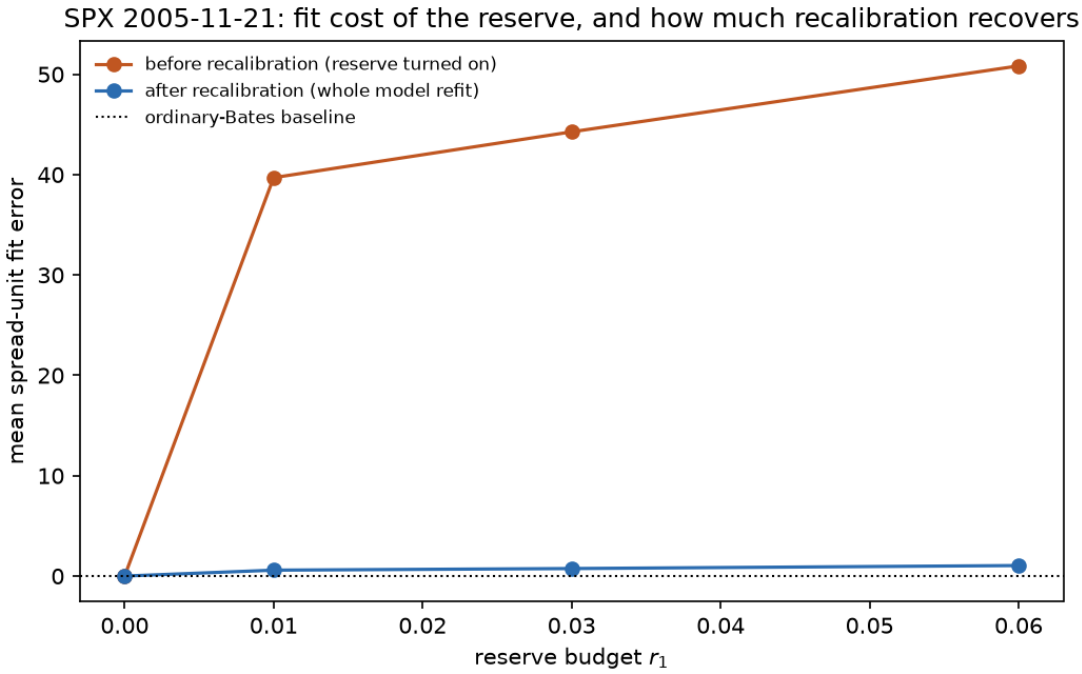


Figure 3: Fit cost of the reserve before recalibration, and how much recalibration recovers, 2005-11-21.

6 A mechanical CRC-routine check on the real 6-date parameter path

Beyond the static, per-day comparison above, a narrower, purely CONSTRUCTIVE question was checked directly on real numbers – not a statistical estimation, so it needs no forecasting power to be meaningful: chaining the 6 real SPX dates' already-calibrated DLV×Bates backgrounds chronologically, does the REALISED change in θ and jump parameters from one date to the next fall inside the reserve-CRC's admissible ("permissible drift") region, and does `apply_reset` preserve protected-option prices exactly on these real numbers? Code: `real_data_study/crc_real_sequence.py` (5/5 tests passing).

Within the standard region, 0 of 5 real transitions are admissible. Using the same conventions as the reserve-sweep study (±0.03 theta band, 1.5×-widened jump box): three transitions fail because the realised θ move is 1.7–3.3× too large for the declared band (all involve multi-year gaps); two fail because the TARGET date's own jump-parameter calibration landed exactly on an optimizer bound ($\lambda_J = 5.0$ on 2024-03-05 and 2024-12-26; $m_J = -0.4999$ on 2025-01-03) – itself a flag that those specific fits are poorly identified, not simply "large."

The mechanism is verified exactly correct on real numbers. A separate sensitivity pass widens each transition's region just enough to admit the realised move (not a claim that this wider region is reasonable – a mechanism check only). Result: `apply_reset` used the exact realised $\Delta\theta$ with zero projection in every case, and protected- option prices matched exactly (to floating-point zero) before vs. after the reset on 4 of 5 transitions (Table 4). The 5th hit a genuine numerical wall: admitting the bound-hitting $m_J = -0.4999$ forces a dominating envelope with $\delta_J \approx 1.65$ (jump log-returns with ∼165% std), which overflows the Carr–Madan damped-contour Fourier pricer at every quadrature setting tested – a real fragility of that pricing method for extremely wide envelopes, reported explicitly rather than silently producing NaN.

Transition	Gap	$\Delta\theta$	θ-widen[†]	Blocked by (standard region)	Protected-price diff[‡]
2005-11-21 → 2010-07-07	4.62y	+0.0988	3.5x	θ move too large	0.0 (n=48)
2010-07-07 → 2011-07-07	1.00y	−0.0519	1.8x	θ move too large	0.0 (n=57)
2011-07-07 → 2024-03-05	12.66y	+0.0274	1.0x	jump params (target λ_J at bound)	0.0 (n=51)
2024-03-05 → 2024-12-26	0.81y	−0.0589	2.1x	θ move too large	0.0 (n=18)
2024-12-26 → 2025-01-03	0.02y	−0.0023	0.1x	jump params (target m_J at bound)	numerical failure[§] (n=3)

[†] Factor by which the ±0.03 theta band would need to widen for admissibility (forced-admissible sensitivity pass; jump-box widening required was 1.0x the standard 1.5x factor in every case, i.e. no extra widening beyond the construction's own target-containment margin). [‡] Maximum absolute price difference on protected contracts (expiry < gap) before vs. after `apply_reset`, once the region was forced wide enough to admit the realised move – 0.0 confirms exact price preservation on real numbers. [§] Carr-Madan Fourier pricing overflowed for every quadrature setting tested; a genuine numerical limit of the dominating-envelope construction when forced to contain an extreme (bound-hitting) target jump mean, not resolved by tuning.

Table 4: The 5 real chronological transitions: realised move, admissibility under the standard region, and the forced-admissible mechanism check.

6.1 More data is needed for this to become a genuine historical-dynamics result

This check is deliberately mechanical, not statistical, and that is both its strength and its limit. It required no forecasting power to be informative – but it also cannot, by itself, establish whether a *properly fitted* physical law and reset-proposal policy would track real SPX dynamics well, because:

- **The theta-band failures look like a time-spacing artefact, fixable by denser data.** The shortest real gap available (2024-12-26 to 2025-01-03, ∼1 week) produced by far the smallest realised $\Delta\theta$ (-0.0023, comfortably inside the band on its own), consistent with θ moving roughly in proportion to elapsed time. Weekly or monthly real data would very plausibly bring most transitions inside a modestly-sized band.
- **The jump-box failures are NOT primarily a spacing issue.** All three bound-hitting calibrations (2024-03-05, 2024-12-26, 2025-01-03) are exactly the dense, modern dates where Sec. 4 already found DLV×Bates fit quality collapsing (13–27% inside bid-ask) – the same underlying model- capacity problem surfacing twice. Sampling MORE modern dates would likely surface more of the same degenerate fits, not fewer. Fixing this needs model/calibration changes (looser or better-justified bounds, more model flexibility, or a joint/warm-started sequential fit across the path instead of independent day-by-day re-calibration), not denser sampling alone.

This environment has only 6 real, non-adjacent option-quote dates (2005–2025, gaps from 7 days to 13 years; see DATA_PROVENANCE.md). That is enough for the cross-sectional comparison above and for the mechanical check in this section, but not enough to fit or validate a genuine historical transition law, physical risk premia, or an OCE/hedging challenger test with any statistical power – at most one real training transition and one held-out forecast point exist in the closest-spaced real subsequence. A full historical-dynamics estimation (latent-state filtering, an explicit P/Q relation with $b^P - b^Q = \Sigma\lambda$, admissible θ-reset proposal dynamics fit jointly across the path, chronological train/validation/held-out testing, and held-out OCE/hedging tests) is specified and ready to run as soon as a denser real series is available – see `real_data_study/HISTORICAL_SIMULATOR_READY_PROMPT.md` for the exact task specification to resume verbatim, and `DATA_PROVENANCE.md`/`DATA_STATUS.md` for exactly what data access (WRDS credentials or a denser pre-populated `QUANTDATA_ROOT`) would unblock it.

7 Outputs and reproducibility

- **Data provenance and exclusions**: `DATA_PROVENANCE.md` and each day's `excluded` list in `report/data/full_study_results.json`.
- **Per-quote predictions and errors**: every evaluated contract's price, spread-unit fit error, and implied-vol error, per model per day, are in `report/data/full_study_results.json` (`by_bucket` maturity/moneyness breakdowns) — not just aggregate statistics.
- **Fitted states, transition specifications, and reserve arrays**: chain κ, anchor grid, Bates background, reserve schedule/box/envelope, and every multi-start diagnostic (initial guess, convergence flag, solver message, final cost) for every model/day, same file.
- **Surface plots**: market quotes vs. dy2 vs. DLV×Bates (Fig. 2); bid-ask coverage by model/date (Fig. 1).
- **Fit-vs-recalibration-freedom plot**: Fig. 3.
- **This PDF report and its LaTeX source**: `report/RealData_CRC_Comparison_Report.{tex,pdf}`.
- **One-command reproduction**:

  ```
  QUANTDATA_TEST_MODE=generate_tests .venv312/bin/python \
  quantdata/dysanos/bates_crc/real_data_study/run_full_study.py && \
  .venv312/bin/python quantdata/dysanos/bates_crc/real_data_-
  study/plots.py
  ```

  run from `quant_data-main/`. This creates/refreshes small derived- cache `.pdq` files under `tests/stored_data/{processing,training}/` (consistent with this repository's existing convention of committing such caches for reproducibility, per `DATA_PROVENANCE.md` Sec. 6).

8 What is established, what is limited, and what remains open

Claim	Status	Basis / caveat
Chain martingale property (any κ, any anchor grid)	**Analytical guarantee**	Exact zero-drift birth-death construction; re-verified post-fit on every calibrated model.
Real-data provenance (12 genuine day-books)	**Verified**	WRDS OptionMetrics via local WIDB fixtures; not dy2.pdf's full historical series.
DLV×Black fits real SPX well	**Refuted**	10.4% inside bid-ask on the selection date; a structural, single-transition-rate limitation, not a bug.
DLV×ordinary-Bates fits real SPX well	**Refuted on modern dense books**	94–100% inside bid-ask on 2005–2011, but 13–27% on 2024–2025 held-out dates – a genuine generalisation gap, not a selection-date artefact (Fig. 1).
Reserve cost is real, large, and only partially recoverable by recalibration	**Verified numerically**	195–276% mean price shift, 0% bid-ask coverage on turning the reserve on; recalibration recovers most but not all on 2/3 dates, fails to converge on the third. Reported as measured changes, not inferred from the admissible interval's width alone.
Node-count sensitivity ($N \in \{5, 7, 9\}$)	**Verified: negligible effect**	Run on 2011-07-07: 92.2%/18.8bp, 94.1%/18.7bp, 92.2%/19.2bp inside bid-ask / median IV err for N=5, 7, 9 – statistically indistinguishable, consistent with the chain component being nearly redundant for this variant (Sec. 2.1, finding 3).
AAPL results	**Secondary only**	Held-out dates only, single-name book, smaller sample; not part of the primary SPX comparison.
CRC reset mechanism correctness on real parameter values	**Verified exactly**	Forced-admissible sensitivity pass: zero projection, exact (0.0) protected-option price preservation on 4/5 real transitions; 5th hits a reported numerical limit, not a bug (Sec. 6).
Real historical transitions admissible in a realistic drift region	**Refuted: 0 of 5**	3/5 blocked by theta-band size (plausibly fixable with denser data); 2/5 blocked by bound-hitting jump calibrations on the same dense-modern dates already found to fit poorly – not fixable by denser sampling alone (Sec. 6.1).
Full multi-year historical dynamics, transition likelihood, physical risk premia	**Out of scope, blocking requirement identified precisely**	Requires a continuous, chronologically regular WRDS/Massive series, unavailable in this environment. Exact resumption task specified in `HISTORICAL_SIMULATOR_READY_PROMPT.md`, ready to paste in as soon as such data is available.

Table 5: Analytical guarantees vs. numerical findings vs. open/limited scope.

9 Conclusion

On the real, if small and non-continuous, WIDB fixture set available in this environment, the coherent DLV×extended-Bates family's answer to "can it fit actual option surfaces well" is nuanced, not a flat yes or no, and the held-out split earns its keep by catching a genuine generalisation gap rather than a design-time artefact:

- The chain-only (Black) variant demonstrably cannot fit real SPX books at all (4–12% inside bid-ask on every date tested), for a structural reason: a single transition rate cannot serve both short- and long-dated smiles at once.
- The Bates-background variant fits close to exactly on the two selection dates and one older held-out date (2005–2011: 94–100% inside bid-ask) – but its own extra chain degrees of freedom are nearly redundant there (the fitted anchor half-width sits at its lower bound). On the three DENSE, MODERN held-out dates (2024–2025) it collapses to 13–27% inside bid-ask (Fig. 1) – a genuine generalisation gap, discovered only because the held-out dates were never used to pick a design choice, exactly the scenario the selection/held-out split exists to expose.
- The reserve extension's fit cost is measured, not inferred: turning on even a modest budget shifts prices 195–276% on average and collapses fit to 0% inside bid-ask; recalibration recovers most (not all) of this on two of three tested dates, and fails to converge outright on the third (2024-03-05 – itself already the weakest ordinary-Bates fit). Once a reserve regime is established, further admissible resets within it are comparatively cheap (one to two orders of magnitude smaller price impact) – establishing the regime is where the real cost lives.

Taken together: the coherent-chain DLV construction, even in its most capable (Bates) form and even after a disciplined two-stage calibration fix caught on a selection date, does not reliably fit today's real, dense SPX book well – and turning on a useful CRC reserve is a materially expensive, only-partially-recoverable regime change, not a free source of scenario diversity. Both are honest, load-bearing findings this task asked for, not shortcomings of the pipeline that built them.

The next bottleneck is data, not methodology, and it is precisely characterised. The CRC reset mechanism itself is exactly correct on real calibrated numbers (Sec. 6) – the open question is not "does the construction work" but "is the available real option history dense and regular enough to fit and validate it as a historical simulator." This environment's 6 non-adjacent real dates answer that question precisely: no. Denser (weekly-or-better) real data would likely resolve the theta-band admissibility failures directly; it would NOT by itself resolve the jump-parameter identifiability failures, which trace to the same modern-book model-capacity gap documented above. A full historical- dynamics estimation (latent-state filtering, an explicit P/Q relation, admissible θ-reset proposal dynamics fit jointly across the path, chronological train/validation/held-out testing, held-out OCE/hedging tests) is specified in full and ready to resume verbatim – `real_data_study/HISTORICAL_SIMULATOR_READY_PROMPT.md` – as soon as either live WRDS access or a denser pre-populated `QUANTDATA_ROOT` becomes available.